

SYNTH-OEM-A diagnostic specification

OEM OEM-A | version 1.0

Data identifiers				
DID	Name	Bytes	Read sessions	Write sessions
0x1000	DID-0	4	1, 3	3
0x1001	DID-1	4	1, 3	3
0x1002	DID-2	4	1, 3	3
0x1003	DID-3	4	1, 3	3
0x1004	DID-4	4	1, 3	3

Data identifiers				
DID	Name	Bytes	Read sessions	Write sessions
0x1005	DID-5	4	1, 3	3
0x1006	DID-6	4	1, 3	3
0x1007	DID-7	4	1, 3	3
0x1008	DID-8	4	1, 3	3
0x1009	DID-9	4	1, 3	3

Prose appendix: operating conditions

DID 0x1000 may be written only when signal_0 is true.

Writing DID 0x1000 requires signal_0 to be true.

DID 0x1000 must not be written while signal_0 is false.

DID 0x1001 may be written only when signal_1 is true.

Writing DID 0x1001 requires signal_1 to be true.

DID 0x1001 must not be written while signal_1 is false.

DID 0x1002 may be written only when signal_2 is true.

Writing DID 0x1002 requires signal_2 to be true.

DID 0x1002 must not be written while signal_2 is false.

DID 0x1003 may be written only when signal_3 is true.

Writing DID 0x1003 requires signal_3 to be true.

DID 0x1003 must not be written while signal_3 is false.

DID 0x1004 may be written only when signal_4 is true.

Writing DID 0x1004 requires signal_4 to be true.

DID 0x1004 must not be written while signal_4 is false.

DID 0x1005 may be written only when signal_5 is true.

Writing DID 0x1005 requires signal_5 to be true.

DID 0x1005 must not be written while signal_5 is false.

The six documented operating conditions are signal_0 through signal_5; each must be true before its corresponding write is accepted.

```
sessions: [{ 'id': 1, 'name': 'default' }, { 'id': 3, 'name': 'extended' }]
```

```
sessions[0].id: 1
```

```
sessions[0].name: "default"
```

```
sessions[1].id: 3
```

```
sessions[1].name: "extended"
```

```
security_levels: [{ 'level': 0, 'name': 'locked' }, { 'level': 1, 'name': 'level1' }]
```

```
security_levels[0].level: 0
security_levels[0].name: "locked"
security_levels[1].level: 1
security_levels[1].name: "level1"

services: [{ 'sid': 16, 'name': 'SessionControl', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],
'suppress_positive_response_supported': False}, { 'sid': 17, 'name': 'Reset', 'subfunctions': [{ 'value': 1, 'name': 'hard' }],
'allowed_sessions': [], 'required_security_level': None, 'suppress_positive_response_supported': False}, { 'sid': 34,
'name': 'ReadDataByIdentifier', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],
'suppress_positive_response_supported': False}, { 'sid': 39, 'name': 'SecurityAccess', 'allowed_sessions': [],
'required_security_level': None, 'subfunctions': [], 'suppress_positive_response_supported': False}, { 'sid': 46, 'name':
'WriteDataByIdentifier', 'allowed_sessions': [], 'required_security_level': None, 'subfunctions': [],
'suppress_positive_response_supported': False}, { 'sid': 49, 'name': 'RoutineControl', 'allowed_sessions': [3],
'required_security_level': 1, 'subfunctions': [], 'suppress_positive_response_supported': False}, { 'sid': 62, 'name':
'TesterPresent', 'suppress_positive_response_supported': True, 'allowed_sessions': [], 'required_security_level': None,
'subfunctions': []}]

services[0].sid: 16
services[0].name: "SessionControl"
services[0].allowed_sessions: []
services[0].required_security_level: null
services[0].subfunctions: []
services[0].suppress_positive_response_supported: false
services[1].sid: 17
services[1].name: "Reset"
services[1].subfunctions: [{"name": "hard", "value": 1}]
services[1].allowed_sessions: []
services[1].required_security_level: null
services[1].suppress_positive_response_supported: false
services[2].sid: 34
services[2].name: "ReadDataByIdentifier"
services[2].allowed_sessions: []
services[2].required_security_level: null
services[2].subfunctions: []
services[2].suppress_positive_response_supported: false
services[3].sid: 39
services[3].name: "SecurityAccess"
services[3].allowed_sessions: []
services[3].required_security_level: null
services[3].subfunctions: []
services[3].suppress_positive_response_supported: false
```

```
services[4].sid: 46
services[4].name: "WriteDataByIdentifier"
services[4].allowed_sessions: []
services[4].required_security_level: null
services[4].subfunctions: []
services[4].suppress_positive_response_supported: false
services[5].sid: 49
services[5].name: "RoutineControl"
services[5].allowed_sessions: [3]
services[5].required_security_level: 1
services[5].subfunctions: []
services[5].suppress_positive_response_supported: false
services[6].sid: 62
services[6].name: "TesterPresent"
services[6].suppress_positive_response_supported: true
services[6].allowed_sessions: []
services[6].required_security_level: null
services[6].subfunctions: []
routines: [{'rid': 8192, 'name': 'Routine-0', 'control_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []}, {'rid': 8193, 'name': 'Routine-1', 'control_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []}, {'rid': 8194, 'name': 'Routine-2', 'control_types': [1, 2, 3], 'sessions': [3], 'security': 1, 'parameter_lengths': {'1': 1, '2': 0, '3': 2}, 'preconditions': []}]
routines[0].rid: 8192
routines[0].name: "Routine-0"
routines[0].control_types: [1, 2, 3]
routines[0].sessions: [3]
routines[0].security: 1
routines[0].parameter_lengths: {"1": 1, "2": 0, "3": 2}
routines[0].preconditions: []
routines[1].rid: 8193
routines[1].name: "Routine-1"
routines[1].control_types: [1, 2, 3]
routines[1].sessions: [3]
routines[1].security: 1
routines[1].parameter_lengths: {"1": 1, "2": 0, "3": 2}
routines[1].preconditions: []
```

```
routines[2].rid: 8194
routines[2].name: "Routine-2"
routines[2].control_types: [1, 2, 3]
routines[2].sessions: [3]
routines[2].security: 1
routines[2].parameter_lengths: {"1": 1, "2": 0, "3": 2}
routines[2].preconditions: []
timing: {'p2_ms': 50, 'p2_star_ms': 5000, 's3_ms': 5000}
nrc_priority: [19, 18, 17, 127, 126, 51, 53, 36, 49, 34]
```